■■ SafeAd AI — Execution & Benchmark Guide

Command Prompt (CMD) Commands, One-Click Scripts, & Video Content Policy Accuracy

Repository: QuadRos-MBCET/main_project_s7 | **Dataset Hub:** safead-dataset-hub | **OS:** Windows

1. One-Click Batch Scripts (.bat)

For instant execution without opening PowerShell or typing long commands, two automated Windows Batch files have been created in the project root directory (c:\Users\hello\Videos\sourav\). You can double-click them in Windows Explorer or run them from CMD:

Script File	Purpose & Execution Behavior
run_video_policy_benchmark.bat	Executes the complete 80-video content policy benchmark against the safead-dataset-hub. Sets paths and prints full accuracy metrics, category breakdown, and confusion matrix.
start_safead_platform.bat	Launches all 3 core services simultaneously in 3 independent CMD windows with health timing: • SafeAd Backend API (Port 8002) • Face-Age API (Port 8001) • Streamlit Unified Web App (Port 8501)

2. Windows Command Prompt (CMD) Execution Guide

Open standard Windows Command Prompt (cmd.exe) and use the exact commands below:

A. Run the 80-Video Content Policy Accuracy Benchmark

This evaluates all 80 videos from the safead-dataset-hub across all 4 policy categories:

```
cd /d c:\Users\hello\Videos\sourav
set PYTHONPATH=c:\Users\hello\Videos\sourav;c:\Users\hello\Videos\sourav\joseph_safead_core
python tests\evaluate_hub_dataset.py
```

B. Start the Complete 3-Service Platform via CMD

Open 3 separate CMD windows (or use start_safead_platform.bat):

Window 1 — SafeAd Backend API (Port 8002):

```
cd /d c:\Users\hello\Videos\sourav
python -m uvicorn backend.app.main:app --host 0.0.0.0 --port 8002
```

Window 2 — Face-Age Estimation API (Port 8001):

```
cd /d c:\Users\hello\Videos\sourav
set PYTHONPATH=c:\Users\hello\Videos\sourav;c:\Users\hello\Videos\sourav\sooraj_face_age_core\face-age
python sooraj_face_age_core\face-age\face_age_api.py
```

Window 3 — Streamlit Web Application (Port 8501):

```
cd /d c:\Users\hello\Videos\sourav
set PYTHONPATH=c:\Users\hello\Videos\sourav
streamlit run frontend/app.py --server.port 8501
```

C. PowerShell Equivalents (One-Liners)

```
cd c:\Users\hello\Videos\sourav
$env:PYTHONPATH='c:\Users\hello\Videos\sourav;c:\Users\hello\Videos\sourav\joseph_safead_core'
python tests/evaluate_hub_dataset.py
```

3. Video Content Policy Accuracy Benchmark Results

The model was evaluated against all 80 authentic video advertisements from **https://github.com/QuadRos-MBCET/safead-dataset-hub**. Below is the empirical performance summary:

Policy Category	Total Videos	Correct	Empirical Accuracy	Target Action
SAFE_18_PLUS	25	23	**92.00%**	RESTRICT (Adult 18+)
SAFE_FOR_ALL	25	22	**88.00%**	APPROVE (Universal)
UNSAFE_FOR_ALL	5	4	**80.00%**	REJECT (Prohibited)
SAFE_14_PLUS	25	15	**60.00%**	RESTRICT (Teens 14+)
OVERALL MULTICLASS	**80**	**64**	**80.00%**	**4-Tier Policy**

Multiclass Confusion Matrix (Actual vs Predicted)

Actual \ Predicted	SAFE_FOR_ALL	SAFE_14_PLUS	SAFE_18_PLUS	UNSAFE_FOR_ALL	Total
SAFE_FOR_ALL	**22**	1	2	0	25
SAFE_14_PLUS	4	**15**	4	2	25
SAFE_18_PLUS	0	0	**23**	2	25
UNSAFE_FOR_ALL	1	0	0	**4**	5
Total Predicted	**27**	**16**	**29**	**8**	**80**

Core Scientific Performance Metrics (Youth Protection Guardrail)

Harmful Ad Interception Recall:	**96.67%** (29 / 30 harmful/adult ads successfully intercepted)
False Acceptance Rate (FAR):	**3.33%** (Only 1 adult ad erroneously passed into youth tiers)
Youth-Safe Content Precision:	**97.67%** (Extremely high purity of youth-approved feed)
False Rejection Rate (FRR):	**16.00%** (Conservative over-flagging on borderline keywords)
Average Video Latency:	**58.9 ms** per video (Ultra-fast real-time inference)

4. Platform Workspaces & User Flows

Workspace / Role	Key Features & Verification Steps
User Workspace (Sooraj Biometrics + Instakill)	1. Upload official College ID card (PNG/JPG/PDF). 2. Auto-DOB extraction with EXIF rotation handling. 3. Start Live Camera feed at locked 30 FPS ArcFace. 4. Click **'Freeze Current Frame'** to cross-match. 5. On match (≥65%), login with username & password to enter Instakill social feed curated to verified age bracket.
Brand Owner Workspace (Ad Campaign Submission)	1. Input creative title and ad caption. 2. Upload video (.mp4) or image ad creative. 3. Click **'Analyze Advertisement'**. 4. Real-time keyframe sampling, OCR, NLP, and policy badge (SAFE FOR ALL, 14+, 18+, or REJECT).
Admin Workspace (Human-in-the-Loop Review)	1. Inspect ads flagged for human moderator audit. 2. View visual safety, OCR, and AI CoT explanation. 3. Issue Override: Approve, Restrict, or Reject.

5. Technical Fixes & Reliability Safeguards

The following production fixes have been implemented in the codebase:

• **Eliminated 0xC0000005 Windows Crash:** Disabled PyTorch MKL-DNN convolutions (torch.backends.mkldnn.enabled = False) to prevent native memory access violations during EasyOCR and ArcFace CPU forward passes.

• **Port Conflict Resolution:** Shifted SafeAd Backend from port 8000 to port 8002 to permanently avoid Windows ephemeral socket collisions with language servers.

• **Camera Session Persistence:** Streamlit session state preserves the Realtime30FPSVisionEngine so uploading or replacing ID cards does not crash the DirectShow camera hardware thread.

• **Mobile EXIF Transpose:** Applied ImageOps.exif_transpose so vertical mobile phone photos of college ID cards are automatically rotated upright before OCR.